

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm Examination

University Theory Examination April/May 2015

PH210 Organic Chemistry-II

Date: 27.04.2015, Monday

Time: 10:00 a.m. to 01:00 p.m.

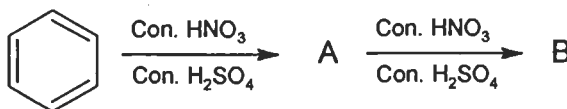
Maximum Marks: 80

Instructions:

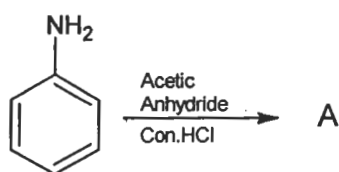
1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

- Q 1** Attempt any TWO of the following; (20)
- A
- i. Explain Electrophilic Aromatic Substitution Reaction with suitable mechanism. 04
 - ii. Explain Fridel-Craft Alkylation and Acylation reaction of benzene with suitable mechanism. 06
- B Write a note on Followings; (2.5 X 04 = 10 Marks) 10
- a. Criteria of Aromaticity
 - b. Neighbouring Group Effect
 - c. Cycloaddition Reaction
 - d. Nano Chemistry
- C
- i. What is "Green Chemistry"? Explain twelve principles of green chemistry in detail. 08
 - ii. **Comment** : Benzene gives substitution reaction while alkene give addition type of reaction. 02
- Q 2** Attempt any TWO of the following (20)
- A
- i. Differentiate Conventional Vs Microwave heating. 05
 - ii. Describe mechanism of sulphonation of benzene. 03
 - iii. Give the answer on following; (01 X 02 = 02 Marks) 02
 - a. Give the structure of A and B.



b. Give the structure of A.



- | | | |
|---|---|----|
| B | i. Explain Sigmatropic reaction with examples. | 05 |
| | ii. Write a brief note on Electrocyclic reaction with suitable example. | 05 |
| C | i. Explain application of Grignard reagent with reaction. | 05 |
| | ii. Preparation and reaction of Amides | 05 |

SECTION- II

- | | | |
|------------|---|------------------------------|
| Q 3 | Attempt any TWO of the following; | (20) |
| A | i. Explain following reactions with mechanism (03x03=09 Marks)
a. Hoffmann Rearrangement for Amide
b. Wolff Kishner Reduction
c. Aldol Condensation

ii. What is Cannizzaro Reaction. | 09

01 |
| B | Give any three preparation of following; (2.5 X 04 = 10 Marks) | 10 |
| | a. Ester
b. Amine
c. Carboxylic Acid
d. Aldehyde | |
| C | i. Give the any two preparation and reaction of Arene.

ii. Give the any two preparation and reaction of Phenol. | 05

05 |
| Q 4 | Attempt any TWO of the following | (20) |
| A | i. Give the any three preparation of Ketone.

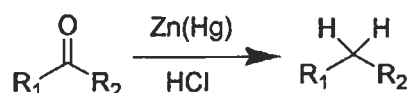
ii. Explain reaction and mechanism of Aceto Acetic Ester Synthesis. | 05

05 |
| B | Answer the following;(Any Two) (02 X 05=10 Marks) | 10 |
| | i. Explain reaction and mechanism of addition of Grignard reagent to ketone.

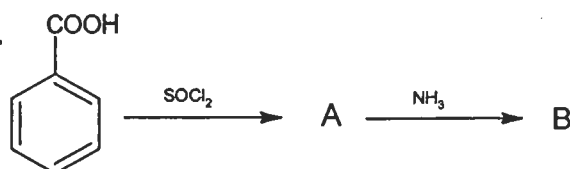
ii. Explain reaction and mechanism of Malonic Ester Synthesis.

iii. Explain Nucleophilic Acyl Substitution Reaction. | |

- i. **Comment:** Carboxylic Acid having higher boiling point than Alcohol.
- ii. **Comment:** Nucleophilic Substitution reaction is takes place much more readily at acyl carbon than saturated carbon.
- iii. **Comment:** Trichloroacetic acid is more acidic than Acetic Acid.
- iv. **Comment:** Aldehydes are more reactive towards Nucleophilic addition reaction than Ketone.
- v. What is Fries Rearrangement? Write a reaction of it.
- vi. Identify the following reactions



- a. Wolff kishner Reduction
 - b. Aldol Condensation
 - c. Reimer Tiemann Reaction
 - d. Clemmensen Reduction
- vii. Betain Shape intermediate found in _____ reaction.
- a. Aceto Acetic Ester synthesis
 - b. Wittig Reaction
 - c. Malonic Ester Synthesis
 - d. Crossed Aldol Synthesis
- viii. What is difference between Aldol and Crossed Aldol Condensation?
- ix. Give the reaction and mechanism of Kolbe Synthesis.
- x. Identify the structure of A and B.



- xi. **Comment:** Cyclopentadiene is not aromatic while anion of Cyclopentadiene is aromatic.